

Answer Key: Molecular Evolution, Mutation, Drift, and Selection

Practice Problem Set

Q1. Field Mice – Fur Color Change

A) It depends on whether the observed change is heritable. If the darker fur is due solely to environmental factors (e.g., temperature or diet), then the population is *not* evolving. However, heritable change can occur without any change in allele frequencies – for example, through inbreeding, which increases homozygosity at a locus with two color-determining alleles, altering the distribution of heritable phenotypes even while allele frequencies remain constant. Therefore, the crucial check is: Is the darkening heritable? The DNA analysis as described (only checking allele frequencies at known fur-color genes) is insufficient to rule out heritable change via altered genotype frequencies or unmeasured loci.

B) No, because adaptation requires a heritable trait that increases fitness; without a genetic basis (or if the change is due to inbreeding without fitness increase), the change is not necessarily an evolutionary adaptation.

C) Possible explanation: Phenotypic plasticity (e.g., darker fur due to environmental factors like temperature or diet) *or* inbreeding increasing homozygosity at a fur-color locus without changing allele frequencies.

Q2. Human Appendix – Adaptation vs. Vestigial Trait

A)

- **Adaptation:** A heritable trait that increases fitness and arose via natural selection.
- **Vestigial trait:** A structure with reduced or no function compared to its ancestral form.

B) Evidence might include: comparative anatomy across species, fitness differences in individuals with vs. without appendix, or functional studies showing immune benefits affect survival/reproduction. If the appendix confers a measurable fitness advantage, it is an adaptation; if it has no effect or a cost with no benefit, it is vestigial.

Q3. Dairy Cow Selection

A) Selection differential

$$S = \text{mean of selected parents} - \text{population mean} = 10,000 - 8,500 = 1,500 \text{ kg}$$

B) Realized heritability

$$h^2 = \frac{R}{S}, \quad R = 9,100 - 8,500 = 600 \text{ kg}$$

$$h^2 = \frac{600}{1500} = 0.4$$

C) Next generation:

$$R = h^2 \times S = 0.4 \times (10,500 - 8,500) = 0.4 \times 2,000 = 800 \text{ kg}$$

$$\text{Predicted new mean} = 8,500 + 800 = 9,300 \text{ kg}$$

Q4. Flower Size Selection

A)

$$R = 4.4 - 4.0 = 0.4 \text{ cm}, \quad S = 5.0 - 4.0 = 1.0 \text{ cm}$$

$$h^2 = \frac{0.4}{1.0} = 0.4$$

B) Next response:

$$R = 0.4 \times (6.0 - 4.0) = 0.4 \times 2.0 = 0.8 \text{ cm}$$

$$\text{Predicted new mean} = 4.0 + 0.8 = 4.8 \text{ cm}$$

Q5. Molecular Clock – Synonymous Differences

Synonymous divergence time = synonymous differences / (2 × substitution rate per site per year)

For X–Y:

$$t = \frac{0.06}{2 \times 1.5 \times 10^{-8}} = \frac{0.06}{3 \times 10^{-8}} = 2 \times 10^6 \text{ years}$$

For X–Z:

$$t = \frac{0.09}{2 \times 1.5 \times 10^{-8}} = \frac{0.09}{3 \times 10^{-8}} = 3 \times 10^6 \text{ years}$$

X and Y share a more recent common ancestor (2 vs. 3 million years).

Q6. dN/dS Ratio = 0.15

A) Purifying (negative) selection.

B) Highly conserved, because most nonsynonymous mutations are deleterious and removed by selection.

Q7. Pseudogene Substitution Rate

A) Rates are similar: 1.2×10^{-8} vs. 1.0×10^{-8} , consistent with neutral theory (most substitutions neutral).

B) If different: small sample error, or very weak selection acting on the pseudogene.

Q8. Fitness Values: AA=1.0, Aa=1.0, aa=0.4

A) The a allele is recessive (fitness effect only in homozygote).

B) Directional selection against aa .

C) Frequency of a will **decrease** because aa has lower fitness.

Q9. Salmon – Recessive Allele d

A) Survival of dd is $100\% - 40\% = 60\%$, so fitness = 0.6. Selection coefficient

$$s = 1 - 0.6 = 0.4$$

B) Before selection: $q_0 = 0.1$, $p_0 = 0.9$

Genotype frequencies: AA = 0.81, Aa = 0.18, aa = 0.01

After selection:

$$AA = 0.81 \times 1.0 = 0.81$$

$$Aa = 0.18 \times 1.0 = 0.18$$

$$aa = 0.01 \times 0.6 = 0.006$$

$$\text{Total} = 0.996$$

$$q_1 = \frac{0.18/2 + 0.006}{0.996} = \frac{0.09 + 0.006}{0.996} = 0.0964$$

Q10. Dominant Deleterious Allele H

A) Mutation-selection balance: $q_{eq} \approx \frac{\mu}{s}$ (for dominant, s = selection against heterozygote). If $s = 1$, $q_{eq} = \mu$. If $\mu = 10^{-5}$, $q_{eq} = 10^{-5}$.

B) Because dominant alleles are exposed to selection in heterozygotes, not just homozygotes, so selection removes them more efficiently.

Q11. Recessive Lethal Allele

A) Equilibrium $q_{eq} = \sqrt{\mu/s}$; $s = 1$, $\mu = 1 \times 10^{-5}$

$$q_{eq} = \sqrt{1 \times 10^{-5}} = 0.00316$$

B) Carrier frequency = $2pq \approx 2q = 0.00632$ (since q small).

Q12. Mutation Equilibrium (No Selection)

A)

$$q_{eq} = \frac{\mu}{\mu + \nu} = \frac{3 \times 10^{-5}}{3 \times 10^{-5} + 1 \times 10^{-5}} = \frac{3}{4} = 0.75$$

B) Approach equilibrium in roughly $1/(\mu + \nu)$ generations $= 1/(4 \times 10^{-5}) = 25,000$ generations conceptually.

Q13. Small Bird Population ($N = 15$)

A) Probability of fixation for a neutral allele = initial frequency = 0.05.

B) Heterozygosity will decrease over time due to genetic drift.

Q14. Fish Population Bottleneck

A) Genetic diversity (heterozygosity, allele number) will be reduced.

B) Less heterozygosity in the recovered population than original, because bottleneck causes loss of rare alleles and reduces heterozygosity even after population recovers.